WELCOME

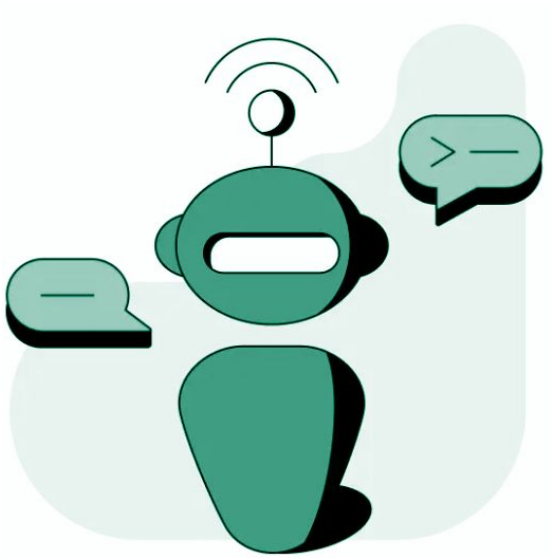
MongoDB x AgenticAI

MongoDB Korea

What is an AI Agent?

AI 에이전트란, 특정 목적을 달성하기 위해서 환경과 업무에 맞게 자율적으로 의사결정을 하고 행동을 취하는 시스템입니다.

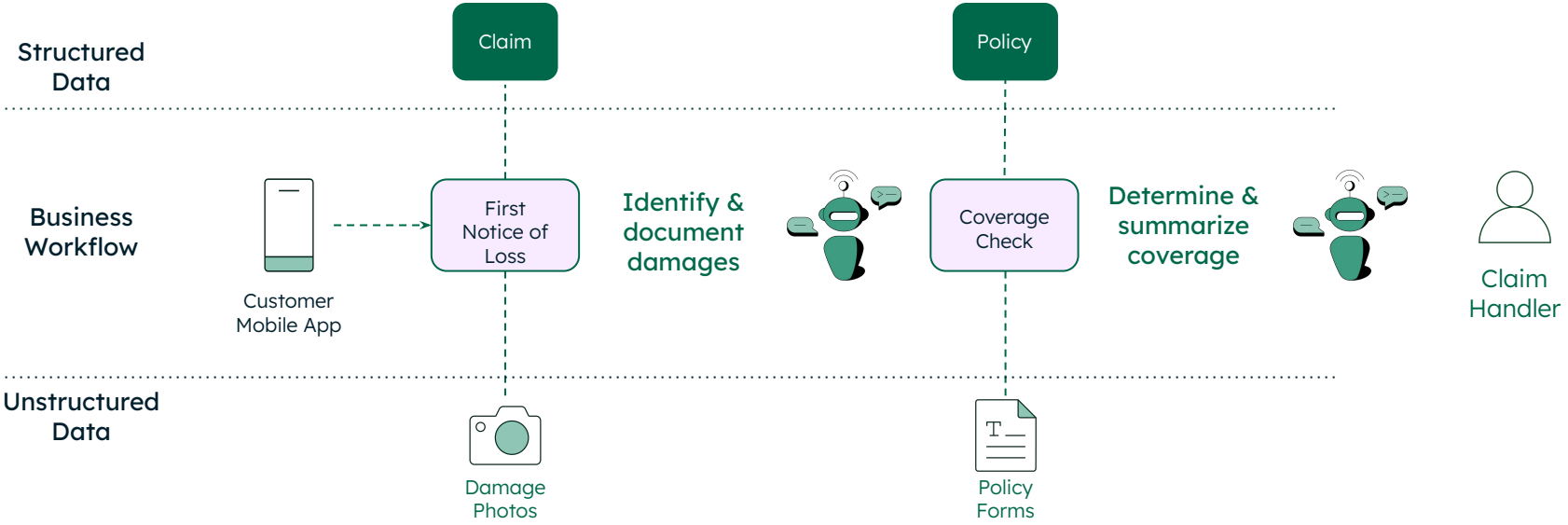
작업 계획, 도구 사용, 검토 및 협업 등과 같은 패턴을 활용합니다.



LLM	AGENT
Chat Prompt Prompted by human	**Autonomous** Can act on its own
Answer One response to question	**Reason** Reason & decide what steps to take
Single Answer Ask questions of the output	**Multi-path** Multiple steps and paths
Stateless Memory limited to session	**Stateful** Long term context & memory
No Tools User input & pre-training data	**Tools** RAG, APIs & Databases

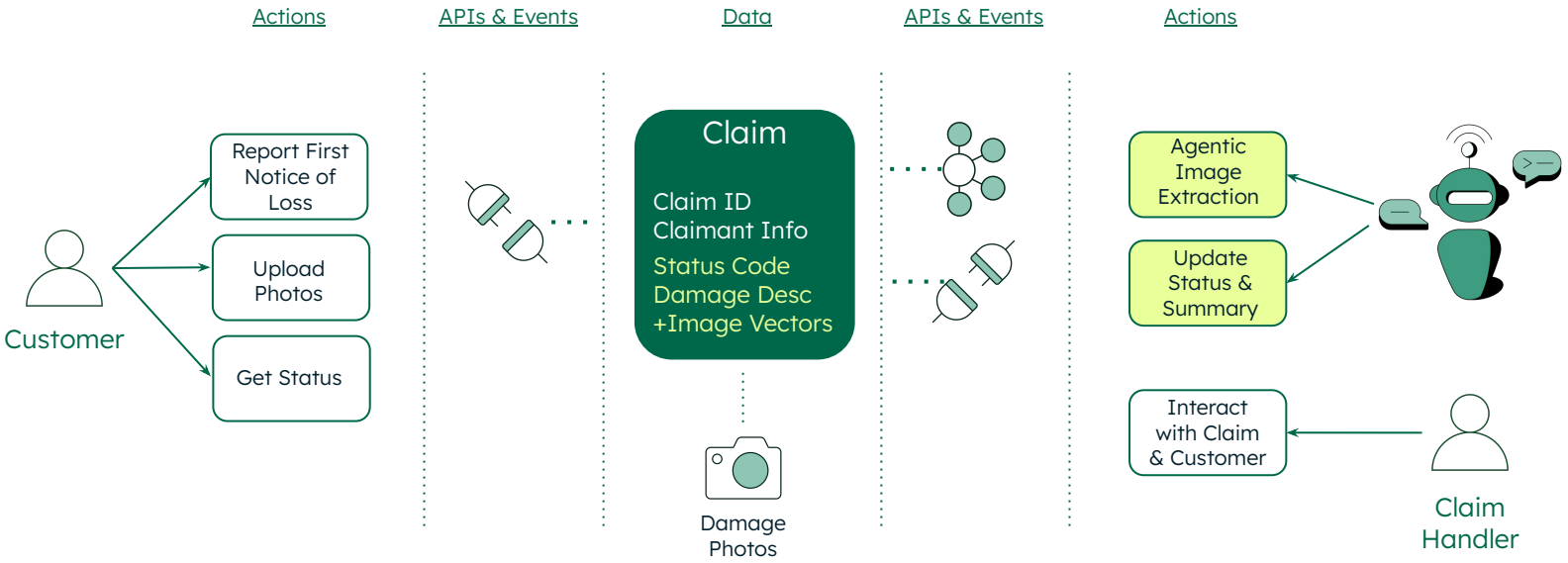
Agentic Workflow

Objective: 운영 비용 절감 및 고객 만족도 향상



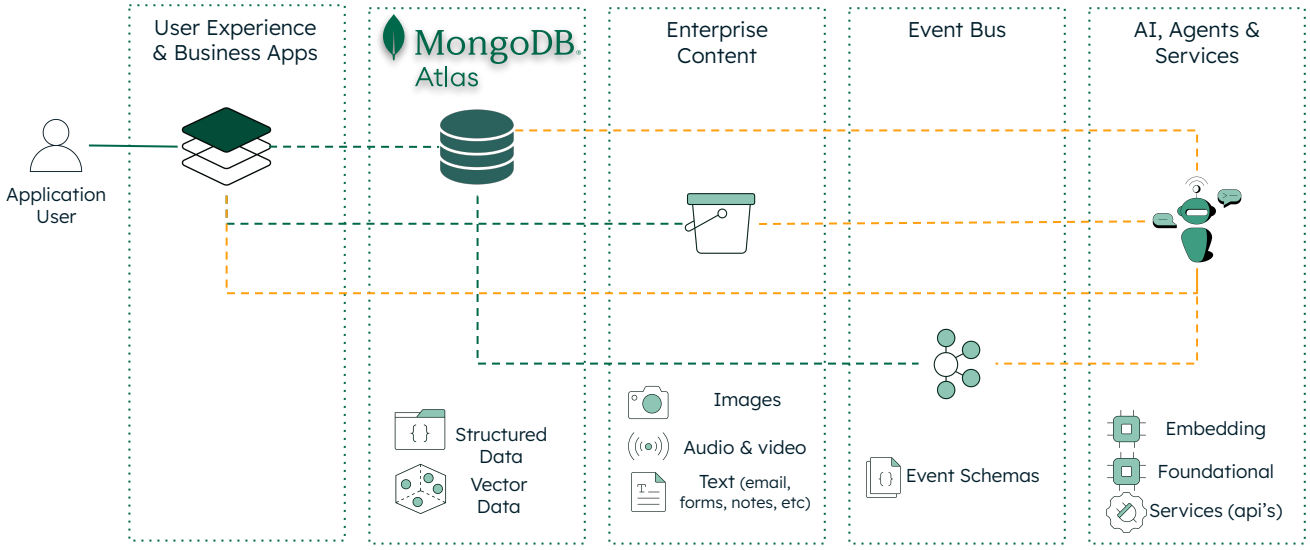
Agentic Architecture

A *Converged* Datastore: APIs & 이벤트, 사용자 & 에이전트, 정형 & 비정형 데이터



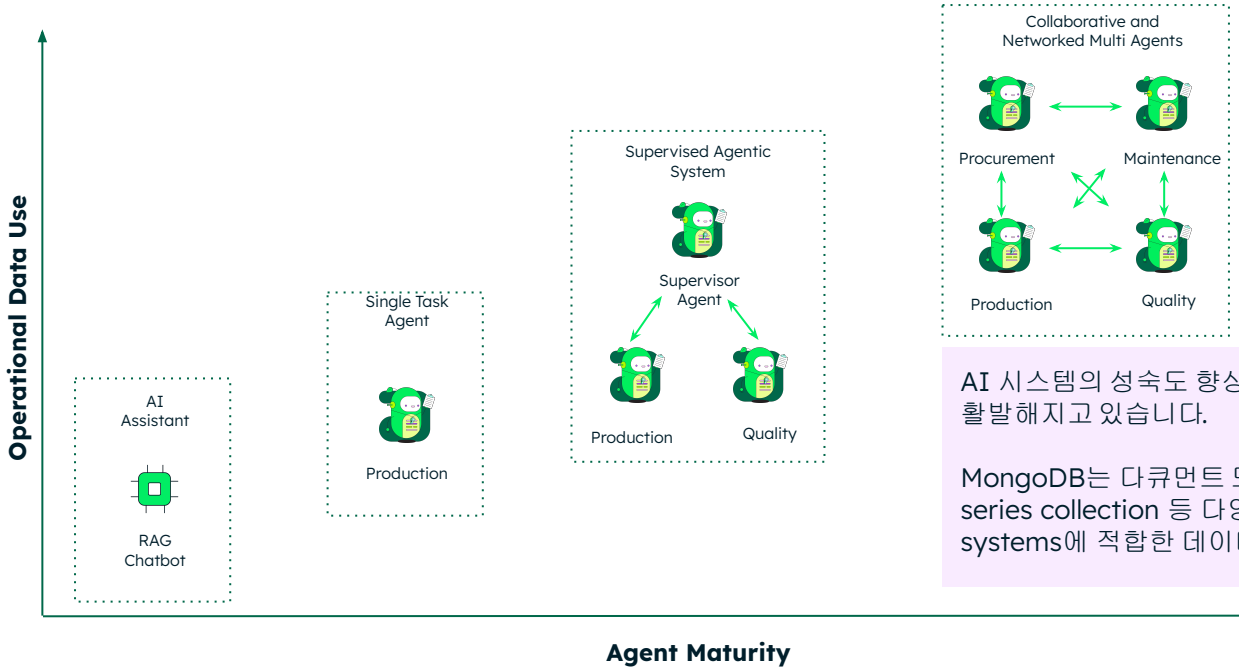
Agentic Architecture

Reference: Online Transactional Processing



transactional

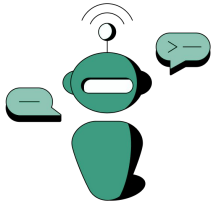
From RAG to Multi-Agent Systems



AI 시스템의 성숙도 향상에 따라 정형 및 비정형 데이터 사용이 더욱 활발해지고 있습니다.

MongoDB는 다큐먼트 모델 기반의 DB로, vector search, time series collection 등 다양한 기능을 제공하고 있어 multi Agentic systems에 적합한 데이터베이스입니다.

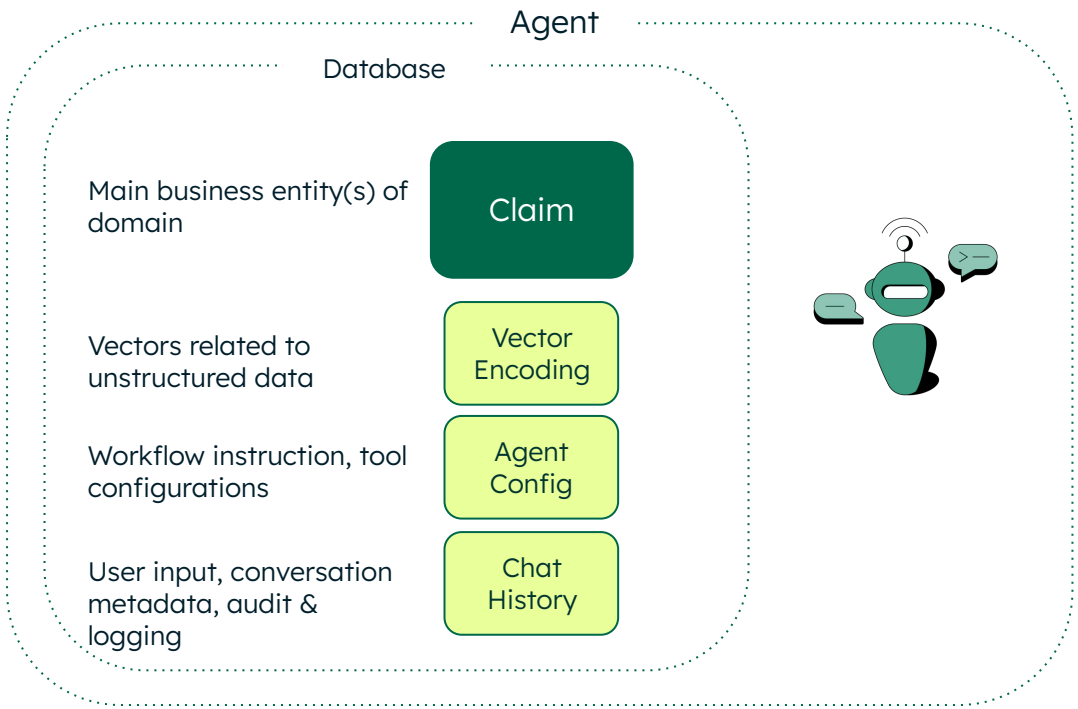
Data that Agents Interact with



AI Agent

- **Perception**
  - Text
  - Audio
  - Images
- **Tools**
  - APIs & events
  - Web Search tools
  - Vector Search
- **Memory**
  - Past Conversations
  - Vectors
  - Business Entities

The Role of a Database in an Agentic Workflow



Why MongoDB for Agentic AI

Document Data Model

Vector Search

Fine Grained Security and Access Controls

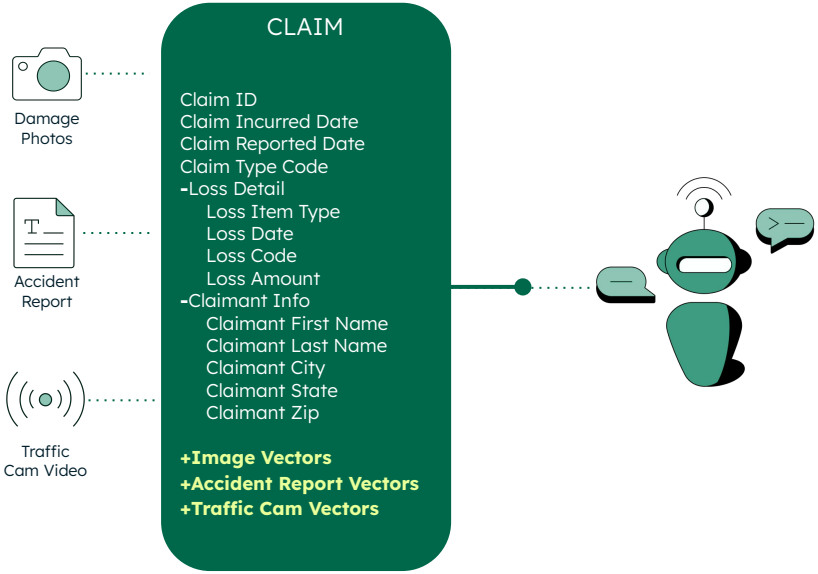
Time Series Collections

MAAP Framework

AI Accelerator

Agentic Architecture

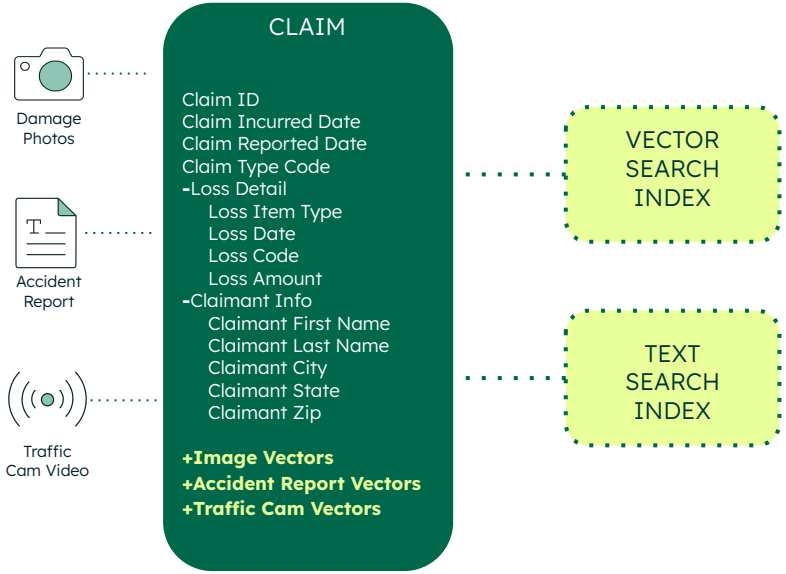
Aggregate Objects for AI



- **JSON** - the AI standard
- 정형 & 비정형 데이터를 효율적으로 저장
- 실시간 처리 가능
- 낮은 지연 시간
  - 대규모 성능

Agentic Architecture

Vector, Text & Hybrid Search

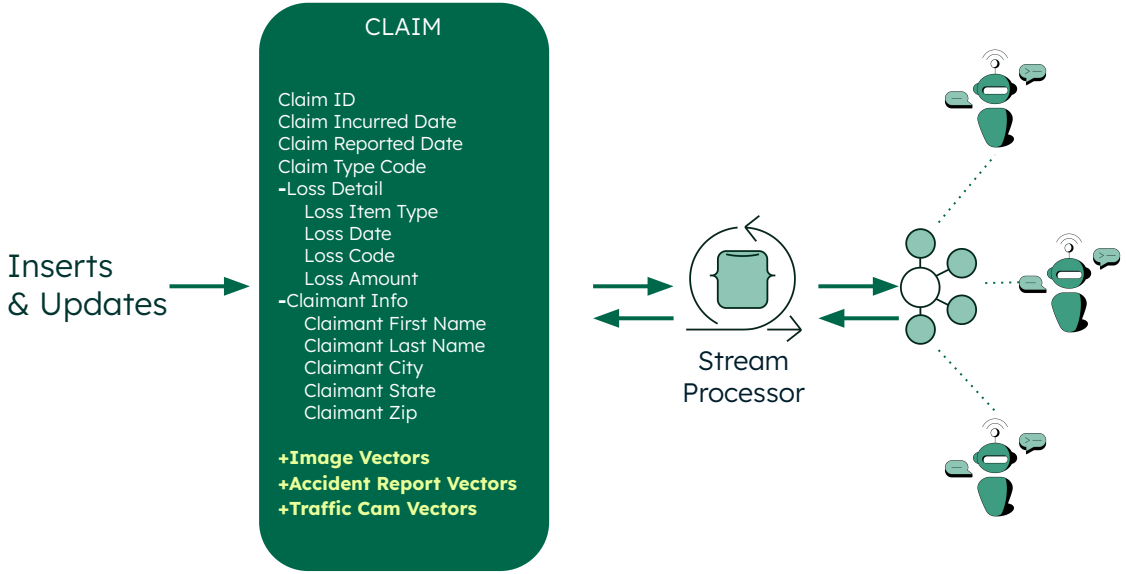


- Vector & Text Search 인덱스 생성으로 쉽게 서치 구현
- Collection 내 변경 사항 자동으로 반영
- $vectorSearch & $search aggregation pipeline 단계 활용해 쉬운 코드 작성
- 아키텍처 장점:
  - 하나의 쿼리 API로 정형&비정형 데이터 활용
  - 낮은 지연시간
  - 낮은 아키텍처 복잡도
  - 높은 보안

Vector Search

Agentic Architecture

Database Eventing - Atlas Stream Processing



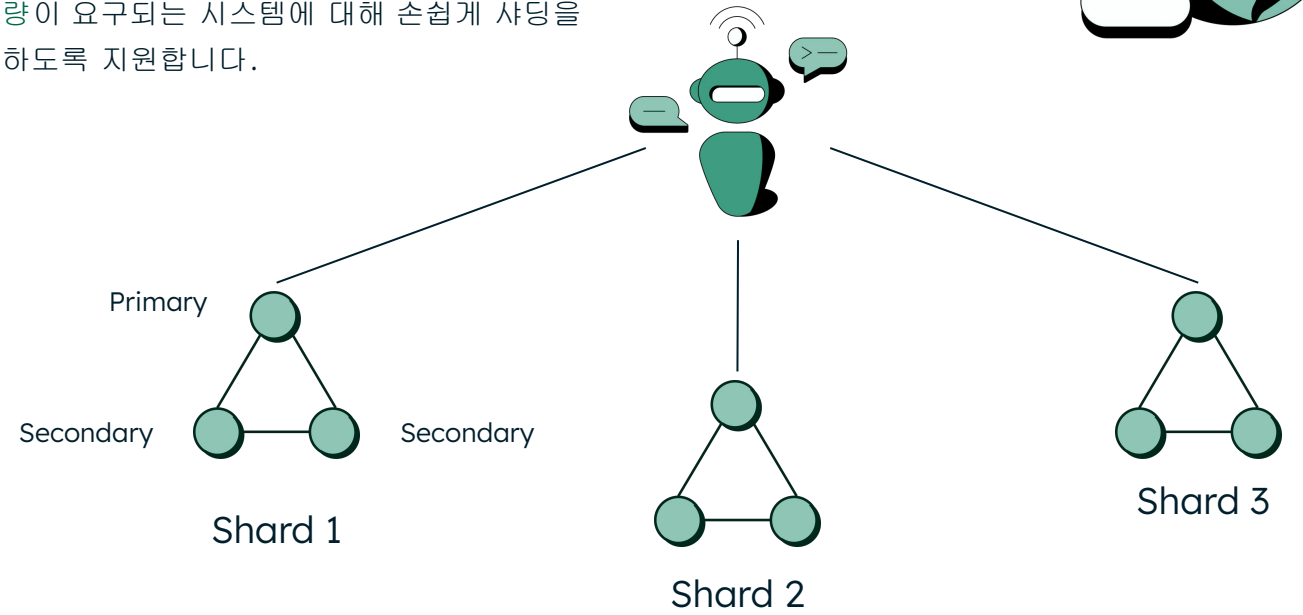
- Atlas의 stream processing 기능을 활용한 실시간 이벤트 처리 가능
- MongoDB Aggregation pipeline으로 스트리밍 데이터 처리
- Atlas로 아키텍처 간소화 및 보안을 충족

Atlas Stream Processing

Agentic Architecture

Sharding: performance at massive scale

샤딩은 여러 서버에 걸쳐 데이터를 분산하는 방식입니다. MongoDB는 대규모 데이터와 높은 처리량이 요구되는 시스템에 대해 손쉽게 샤딩을 구현하도록 지원합니다.



Sharded Clusters

What can we help you build next?